Smart Warehouse WMS: Project Handoff Summary

Version 2.0 · 28 October 2026 · Reflects the design after the late-October change of plans

1. The project

Senior capstone at Fathalla International Applied Technology School: the Smart Warehouse Management and Pick-to-Light System. Three parts talk to one backend: an Android worker app, an admin dashboard on PC (ASP.NET MVC), and ESP32/Arduino controllers that drive LEDs on the shelves. The backend is an ASP.NET Core Web API with EF Core and SQL Server.

2. Roles

- Admin (one per warehouse): enters orders, chooses the shelf and the worker for every order line, assigns shelves to inbound stock, monitors inventory.
- Worker: logs in by email, receives assigned orders, scans and picks, follows the map route, confirms handoff, and creates inbound records.

3. How it works now

- Outbound: the admin creates an order with lines on specific shelves and assigns each line to a worker. Stock is committed immediately and the workers are notified. Each pick scan lowers shelf stock. After a worker's last pick the app routes them to the loading dock, and they confirm handoff; the order completes when every line is handed off.
- Inbound: the worker scans the incoming product's barcode and enters cartons and the voucher number in the Android app. The admin assigns a shelf (the designated shelf, otherwise the closest empty one). The worker places the cartons and confirms.
- Pick-to-light: the ESP32 controllers poll the backend and light the LED of the shelf to pick from.
- Floor map: an exported map image plus a graph held on the server. The worker selects the shelf they are standing next to, and the server returns the path to the next shelf and finally to the dock.

- Stock: each shelf has OnHand and Committed. Available is the difference, and new orders are checked against it.

4. Decisions log

Topic	Decision
Concurrency	No shelf locking or reservation. Orders run in parallel.
Worker assignment	Manual, by the admin from the dashboard, one worker per order line.
Availability	OnHand minus Committed, per shelf.
Order creation	Validate all lines, then write all of them in one transaction with conditional updates. Roll back if any line fails.
Releasing committed stock	Per line at each confirmed pick, together with the on-hand deduction.
Cancel	Only before the first confirmed pick.
Admins	One per warehouse, stated as an explicit assumption.
Shelf rule	One product per shelf. When the designated shelf is full, use the closest empty shelf.
Pick-source shelves	Chosen by the admin when creating the order.
Worker position	The worker selects the shelf they are standing next to.
Routing	On the server, Dijkstra, fixed rack-then-level visiting order.
Dashboard updates	Polling, no SignalR.
Hardware link	ESP32 polls the backend over HTTP.
Database	SQL Server.
Branch order submission	Out of scope for now. The admin enters orders.

5. Removed from the earlier design

Shelf-level locking, the reservations table, the waiting queue with retry-on-completion, FIFO ordering of waiting orders, the starvation alert, the priority hold, the rule against overlapping stock between orders, automatic load balancing (fewest orders, logged-in workers only), and the rule against live indoor navigation.

6. Team split

- Person A: server and hardware.
- Person B: Android app and admin dashboard.

7. Review-day demo

A laptop running the admin dashboard, an Android phone running the worker app, a display screen, and a wooden scale model with 4 to 5 shelf levels, each with an LED wired to a microcontroller that polls the backend.

8. Documents

- ER Schema and API Specification v2.0: authoritative for tables and endpoints.
- Mission: Smart Warehouse Core API v2.0: the backend build brief.
- This summary.

Smart Warehouse WMS: Design Document v1.0 is written and covers the full capstone design.

9. Open items and confirmations

1. Confirmed: handoff is per worker per order, and the order completes when all lines are handed off.
2. Confirmed: the inbound lifecycle (stock is counted at placement, and the admin assigns the shelf).
3. Deferred until the documentation is final and both agents know the project: measuring node coordinates on the exported map image, seeding the graph, and checking that every shelf anchor and the dock are connected.
4. Done: the one-admin assumption is written into the Design Document (section 3.3).